

Question	Marking Guidance	Mark	Comments
08.1	1. (Requires DNA fragment) DNA polymerase, (DNA) nucleotides and primers; 2. Heat to 95 °C to break hydrogen bonds (and separate strands); 3. Reduce temperature so primers bind to DNA/strands; 4. Increase temperature, DNA polymerase joins nucleotides (and repeat method);	4	1 and 4. Accept Taq polymerase for DNA polymerase. 2. Accept temperature in range 90 to 95 °C. 3. Accept temperature in range 40 to 65 °C. 4. Accept temperature in range 70 to 75 °C.
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08.2	1. (Initially) number (of molecules) doubling is low OR Doubles each cycle to produce exponential increase; 2. Plateaus as no more nucleotides/primers;	2	1. First alternative relates to idea of low numbers i.e., 2, 4, 8, 16, 32 etc. 2. Accept 'levels out' or 'flattens' for plateaus. 2. Accept enzyme/polymerase (eventually) denatures.

